

ANX-PR/CL/001-01

LEARNING GUIDE

SUBJECT

203000035 - Big Data Engineering

DEGREE PROGRAMME

20BC - Master Universitario En Biología Computacional

ACADEMIC YEAR & SEMESTER

2025/26 - Semester 1

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1. Description

1.1. Subject details

Name of the subject	203000035 - Big Data Engineering
No of credits	3 ECTS
Type	Optional/elective
Academic year of the programme	First year
Semester of tuition	Semester 1
Tuition period	September-January
Tuition languages	English
Degree programme	20BC - Master Universitario en Biología Computacional
Centre	20 - Etsi Agronómica, Aliment. Y Biosistemas
Academic year	2025-26

2. Faculty

2.1. Faculty members with subject teaching role

Name and surname	Office/Room	Email	Tutoring hours *
Jose Manuel Moya Fernandez (Subject coordinator)		jm.moya@upm.es	- -

* The tutoring schedule is indicative and subject to possible changes. Please check tutoring times with the faculty member in charge.

3. Prior knowledge recommended to take the subject

3.1. Recommended (passed) subjects

The subject - recommended (passed), are not defined.

3.2. Other recommended learning outcomes

- Python programming

4. Skills and learning outcomes *

4.1. Skills to be learned

CE02 - Utilizar sistemas operativos, programas y herramientas de uso común en biología computacional, así como, manejar plataformas de cómputo de altas prestaciones, lenguajes de programación y análisis bioinformáticos

CE03 - Analizar e interpretar bioinformáticamente los datos que se derivan de las tecnologías ómicas, y proponer soluciones bioinformáticas en relación a dichos datos.

CE05 - Utilizar herramientas de biología computacional para el análisis genómico, incluida la genómica comparativa y biología evolutiva.

CE10 - Conocimiento de las técnicas de representación del conocimiento reutilizables y modelos de razonamiento en entornos centralizados y distribuidos a utilizar en la resolución de problemas que impliquen conducta inteligente.

CG01 - Poseer los conocimientos que constituyen la base científica y tecnológica de la Biología computacional, lo que permitirá el desarrollo de ideas originales en este campo, en un contexto de investigación o desarrollo.

CG03 - Que los estudiantes sepan aplicar los conocimientos adquiridos y su capacidad de resolución de problemas en entornos nuevos o poco conocidos dentro de contextos más amplios (o multidisciplinares) relacionados con el área de la Biología Computacional.

CT08 - Tener capacidad de análisis y síntesis para interpretar datos relevantes y abordar los problemas desde diferentes perspectivas.

4.2. Learning outcomes

RA48 - Entender qué es el análisis de grandes volúmenes de datos, sus características, los métodos y las plataformas más importantes

RA49 - Entender la plataforma Apache Spark, sus objetivos, sus componentes y su modelo de programación

RA51 - Utilizar Spark para desarrollar productos y servicios intensivos en datos, como sistemas de recomendación, predicción y diagnóstico, utilizando procesamiento paralelo, Spark Machine Learning Pipelines y Spark Streaming

RA52 - Utilizar SparkR para realizar análisis estadístico en grandes volúmenes de datos.

RA50 - Escribir y depurar aplicaciones Spark para el análisis de datos masivos

* The Learning Guides should reflect the Skills and Learning Outcomes in the same way as indicated in the Degree Verification Memory. For this reason, they have not been translated into English and appear in Spanish.

5. Brief description of the subject and syllabus

5.1. Brief description of the subject

The analysis of large volumes of data is used to support decision making and to build data intensive products and services, such as recommendation, prediction and diagnostic systems. The collection of skills required to perform these functions have been grouped under the term Data Science. This course develops the necessary skills around the Apache Spark platform, which is rapidly becoming the reference platform for mass data analysis. The course is structured around practical projects that include prediction, using automatic learning algorithms, collaborative filtering and pattern search. These exercises demonstrate in a practical way how to manipulate data sets through parallel processing with PySpark, Spark SQL, Spark Machine Learning Pipelines and SparkR.

In this course, you will:

- Understand what high-volume data analysis is, its features, methods, and major platforms.
- Understand the Apache Spark platform, its objectives, its components, and its programming model.
- Write and debug Spark applications for mass data analysis.
- Use Spark to develop data-intensive products and services, such as recommendation, prediction, and diagnostic systems, using parallel processing, Spark Machine Learning Pipelines, and Spark Streaming.
- Use PySpark perform statistical analysis on large volumes of data

5.2. Syllabus

1. Introduction

1.1. Understanding the Data Lake landscape

1.2. Introduction to Spark

1.3. Setting up Python with Spark

2. Spark Dataframe Basics

3. Exploratory Data Analysis with Spark

4. Spark ML Introduction

5. Linear Regression

6. Logistic Regression

7. Tree Methods

8. K-Means Clustering

9. Recommender Systems

10. Natural Language Processing

11. Spark Streaming

12. Scalable Neural Networks

6. Schedule

6.1. Subject schedule*

Week	Type 1 activities	Type 2 activities	Distant / On-line	Assessment activities
1	Introduction Duration: 01:30 Lecture Setting up Python with Spark Duration: 00:30 Laboratory assignments			
2	Spark Dataframe Basics Duration: 01:00 Lecture Spark Dataframe Basics - Exercise Duration: 01:00 Laboratory assignments			Consulting project - Dataframe Basics Individual work Progressive assessment Not Presential Duration: 00:00
3	Exploratory Data Analysis with Spark Duration: 01:00 Lecture Exploratory Data Analysis with Spark - Exercise Duration: 01:00 Laboratory assignments			
4	Spark ML Intro Duration: 01:00 Lecture Spark ML Intro - Exercise Duration: 01:00 Laboratory assignments			
5	Linear Regression Duration: 01:00 Lecture Linear Regression - Exercise Duration: 01:00 Laboratory assignments			Consulting project - Linear regression Individual work Progressive assessment Not Presential Duration: 00:00
6	Logistic Regression Duration: 01:00 Lecture Logistic Regression - Exercise Duration: 01:00 Laboratory assignments			Consulting project - Logistic regression Individual work Progressive assessment Not Presential Duration: 00:00

7	Tree Methods Duration: 01:00 Lecture Tree Methods - Exercise Duration: 01:00 Laboratory assignments			Consulting project - Tree Methods Individual work Progressive assessment Not Presential Duration: 00:00 Challenge Individual work Progressive assessment Not Presential Duration: 00:00
8	K-Means Clustering Duration: 01:00 Lecture K-Means Clustering - Exercise Duration: 01:00 Laboratory assignments			Consulting project - Clustering Individual work Progressive assessment Not Presential Duration: 00:00
9	Recommender Systems Duration: 01:00 Lecture Recommender Systems - Exercise Duration: 01:00 Laboratory assignments			
10	Natural Language Processing Duration: 01:00 Lecture Natural Language Processing - Exercise Duration: 01:00 Laboratory assignments			
11	Spark Streaming Duration: 01:00 Lecture Spark Streaming - Exercise Duration: 01:00 Laboratory assignments			
12	Scalable Neural Networks Duration: 01:00 Lecture Scalable Neural Networks - Exercise Duration: 01:00 Laboratory assignments			
13	Final project Duration: 02:00 Cooperative activities			
14	Final project Duration: 02:00 Cooperative activities			Final project Group work Progressive assessment Not Presential Duration: 00:00

15				
16				
17				Project Individual work Global examination Presential Duration: 04:00

Depending on the programme study plan, total values will be calculated according to the ECTS credit unit as 26/27 hours of student face-to-face contact and independent study time.

7. Activities and assessment criteria

7.1. Assessment activities

7.1.1. Assessment

Week	Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
2	Consulting project - Dataframe Basics	Individual work	No Presential	00:00	15%	3 / 10	CG01 CT08 CE03 CE10 CG03 CE02
5	Consulting project - Linear regression	Individual work	No Presential	00:00	15%	3 / 10	CG01 CT08 CE03 CE10 CG03 CE02
6	Consulting project - Logistic regression	Individual work	No Presential	00:00	15%	3 / 10	CG01 CT08 CE03 CE10 CG03 CE02
7	Consulting project - Tree Methods	Individual work	No Presential	00:00	15%	3 / 10	CG01 CT08 CE03 CE10 CG03 CE02
7	Challenge	Individual work	No Presential	00:00	10%	0 / 10	CG01 CT08 CE03 CE10 CG03 CE02
8	Consulting project - Clustering	Individual work	No Presential	00:00	15%	3 / 10	CG01 CT08 CE03 CE10 CG03 CE02

14	Final project	Group work	No Presential	00:00	25%	3 / 10	CG01 CT08 CE03 CE10 CG03 CE02
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7.1.2. Global examination

Week	Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
17	Project	Individual work	Face-to-face	04:00	100%	5 / 10	CG01 CT08 CE03 CE10 CG03 CE02

7.1.3. Referred (re-sit) examination

Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
Project	Individual work	Face-to-face	04:00	100%	5 / 10	CG01 CT08 CE03 CE10 CG03 CE02

7.2. Assessment criteria

The assessment will check whether students have acquired the competences of the subject. Therefore, the assessment in the ordinary and extraordinary sessions will use the same types of assessment techniques.

ORDINARY SESSION: Students will be assessed using the progressive assessment system which will consist of the following activities:

Individual consulting projects: Practical work on data analysis problems with Spark accounting for 75% of the final mark. An additional grade-raising challenge is included, which allows you to add up to one extra point to your final grade.

Final team project: Practical application of the knowledge acquired to a real-world problem, with minimal guiding from the teaching staff.

The total weight of the evaluation activities is 110%. In case the average mark exceeds 10, it will be truncated to 10.

EXTRAORDINARY SESSION: the evaluation of the subject in its extraordinary session will consist of the following activities:

Final individual project: Practical application on Spark of the knowledge acquired to a real-world problem, with minimal guidance from the teaching staff (classroom activity).

Rationale for a 110 % weighting in progressive assessment

In the continuous-assessment pathway, students complete a large number of graded tasks. While this variety provides ongoing feedback and encourages steady engagement, it also increases the statistical likelihood that even an outstanding student will lose a few points on one component?perhaps through a minor error, illness, or timetable clash. Under a strict 100 % cap, those small deductions can prevent high-performing students from demonstrating the same level of excellence (a full 10/10) that they could achieve through the single final-exam route.

To restore parity, the progressive scheme includes one additional, low-stakes enrichment activity worth 10 % of the

course mark. The cumulative weighting therefore totals 110 %, but the final recorded mark is still capped at 100 %. This extra opportunity serves two purposes:

- **Fairness:** It compensates for the greater exposure to random losses inherent in multiple assessments, ensuring that continuous-assessment students are not disadvantaged compared with those opting for a single exam.
- **Excellence and motivation:** It rewards sustained effort and allows truly exceptional students to offset minor slips and still attain the maximum grade.

In short, the extra 10 % balances the rigor of progressive evaluation with equitable access to top grades.

8. Teaching resources

8.1. Teaching resources for the subject

Name	Type	Notes
Spark Python API documentation	Web resource	
Machine Learning Library (MLlib) Guide	Web resource	
Spark SQL Language Reference	Web resource	
Companion book: An Introduction to Statistical Learning with Applications in R, by Gareth James, Daniela Witten, Trevor Hastie and Robert Tibshirani	Web resource	
Companion book: Jannach, Zanker, Felfernig, Friedrich. Recommender Systems: An Introduction. Cambridge University Press, New York 2011	Web resource	
Google Colab Notebooks	Web resource	
Moodle site	Web resource	
Slides and sample notebooks	Web resource	

9. Other information

9.1. Other information about the subject

This course is related to SDG9, SDG11 and SDG12.